van Emde Boas trees

Part V Advanced Data Structures

Scope

Splitting integer high/low bits supports fast predecessors and successors. Cached min/max makes the doubly logarithmic bound possible by allowing only one expensive recursion per level. Independently stored minima and linear-universe allocation determine update rules and applicability.

Prerequisites

Integer universes, bit splitting, and recursive summaries.

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Chapter route

Learning objectives

- Compare bitmaps, summaries, and clustering.
- Define recursive cluster and summary universes.
- Derive insertion, successor, and min/max costs.
- 20.1 Preliminary approaches
- 20.2 A recursive structure
- 20.3 The van Emde Boas tree

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Preliminary approaches

The integer universe is $U=\{0,\ldots,u-1\}$, $u=2^b$, $b \geq 1$. successor(x) is the smallest stored value **strictly greater** than x; predecessor is strictly smaller. x need not be stored. Return NIL when absent.

Assume word-RAM with word length $w \geq b$. Legal keys, shifts, masks, and array indexing each cost $O(1)$. This is not bit complexity; keys spanning multiple machine words do not inherit the same constant cost.

A u-bit bitmap supports direct membership. Successor may scan forward; even with a fast first-one operation per word, worst case is $\Theta(u/w)$ words. $O(1)$ requires the entire universe to fit one word and the model to provide the necessary bit scan. A red-black tree instead uses $O(n)$ nodes and $O(\log(n+1))$ time on arbitrary comparison keys without allocating u.

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Clustering alone does not imply double logarithms

Split u=256 into16 groups of16 keys. Summary bit i indicates nonempty group i. Successor checks for a larger value in the current group, then finds the next nonempty group and its minimum.

Bit-by-bit scans still cost up to16 within-group steps plus16 summary steps. Group width m gives O(m+u/m), minimized near$\sqrt{}$u at O($\sqrt{}$u), not O(log log u).

Recursing over many levels also requires counting expensive calls per level. The proto structure below shows why taking square roots of the universe alone is insufficient.

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A recursive structure

Split b bits into upper ceil(b/2) and lower floor(b/2) bits:

```
upper=2^ceil(b/2); lower=2^floor(b/2)
high(x)=floor(x/lower); low(x)=x mod lower
index(h,l)=h*lower+l
```

There are upper clusters, each with universe lower. The summary has universe upper and stores nonempty cluster indices. These two sizes cannot always both be written as$\sqrt{u}$.

At u=32, b=5, upper=8, lower=4. Key27 has high6, low3, and recombines as$6 \cdot 4+3=27$. Rounding$\sqrt{32}$ for both parts would lose a complete nonoverlapping bit partition.

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Why proto-vEB's two recursions are slower

First let b be a power of two so splitting is exact. Without cached extrema, proto-minimum recursively finds the first nonempty group in summary and then its local minimum:

$T_{min}(u)=2T_{min}(\sqrt{u})+\Theta(1)$.

With S(b)=T(2^b), S(b)=2S(b/2)+Θ(1)=Θ(b), giving Θ(log u). Insertion that always recurses into both cluster and summary has the same bound.

Proto-successor may search its own cluster unsuccessfully, recurse into summary, then find the next group's minimum:

$T_{succ}(u)=2T_{succ}(\sqrt{u})+\Theta(\log u)$.

Substitution gives S(b)=2S(b/2)+Θ(b)=Θ(b log b), or Θ(log u log log u). Improvement must remove expensive branches, not merely count recursion depth.

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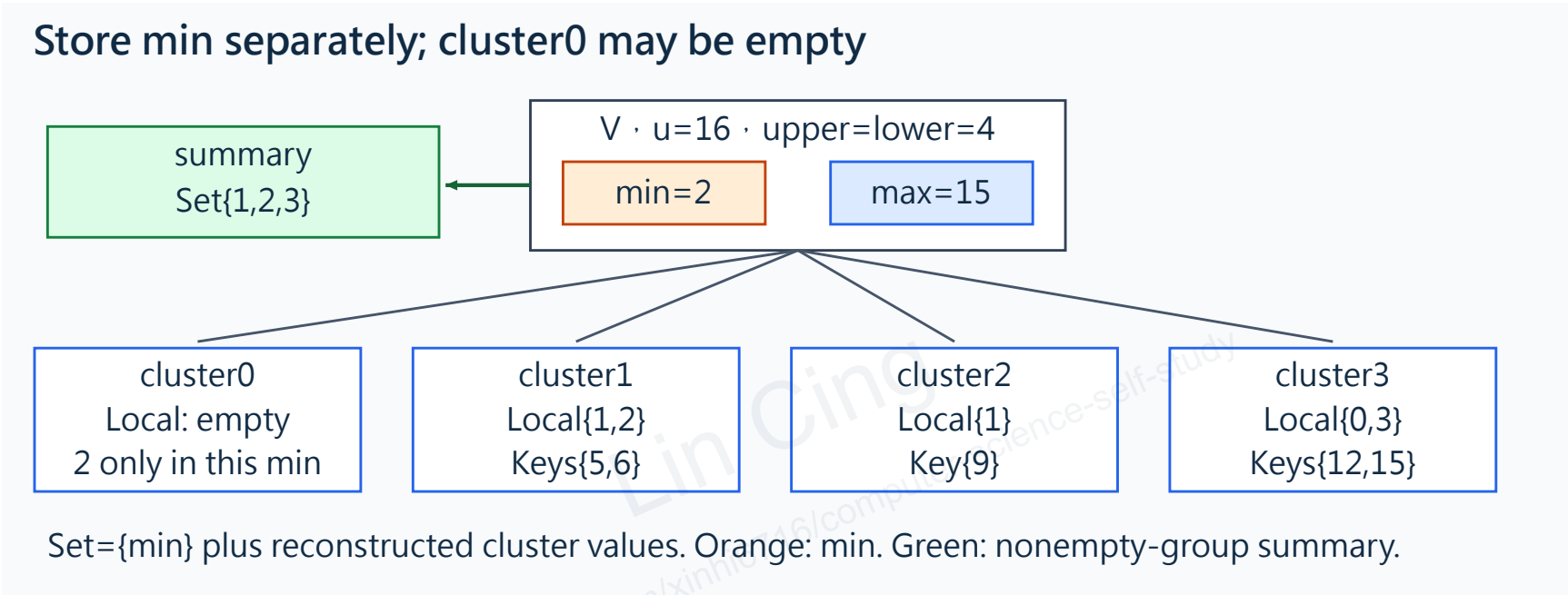
The van Emde Boas representation

Each node stores min,max, both NIL when empty. **min is excluded from all clusters.** Every other key occurs uniquely by high/low in a cluster. Summary contains exactly indices of those nonempty clusters.

A singleton has min=max with no nonempty clusters. With at least two values, max is also stored in a cluster. Base universe u=2 uses only min/max, without further clusters.

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The van Emde Boas representation (continued)



At u=16 and S={2,5,6,9,12,15}, min2,max15; cluster0 empty,cluster1 local{1,2},cluster2{1},cluster3{0,3},summary{1,2,3}. high(2)=0 does not force summary to contain0 because2 exists only in this node's min field.

Empty, singleton, and membership

```
member(V,x):
    if V.min==NIL: return false
    if x==V.min or x==V.max: return true
    if V.u==2: return false
    return member(V.cluster[high(x)],low(x))
minimum(V): return V.min
maximum(V): return V.max
empty-insert(V,x): V.min=V.max=x
```

Membership enters one cluster per level with strictly smaller universe. Extrema and empty/singleton tests cost O(1). empty-insert does not recursively populate the whole structure, since a new min belongs to no cluster. This constant-time base also leaves only one expensive update recursion.

The public interface validates integer $x \in 0..u-1$. It may detect duplicate insertion or absent deletion using member, return false, and leave the set unchanged. Adding one same-order query does not change asymptotic time. Internal update pseudocode assumes a new key or a present key respectively.

Mutually exclusive successor branches

```
successor(V,x):
    if V.min==NIL or x>=V.max: return NIL
    if x<V.min: return V.min
    if V.u==2: return 1
    h=high(x); l=low(x); C=V.cluster[h]
    if C.max!=NIL and l<C.max:
        return index(h,successor(C,l))
    j=successor(V.summary,h)
    if j==NIL: return NIL
    return index(j,V.cluster[j].min)
```

Reading C.max determines whether a larger same-cluster value exists. If so, recurse only there. Otherwise recurse directly into summary without first searching a cluster known to fail. Reading the next cluster's min costs O(1), so there is at most one expensive call per level.

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Predecessor must restore the separate minimum

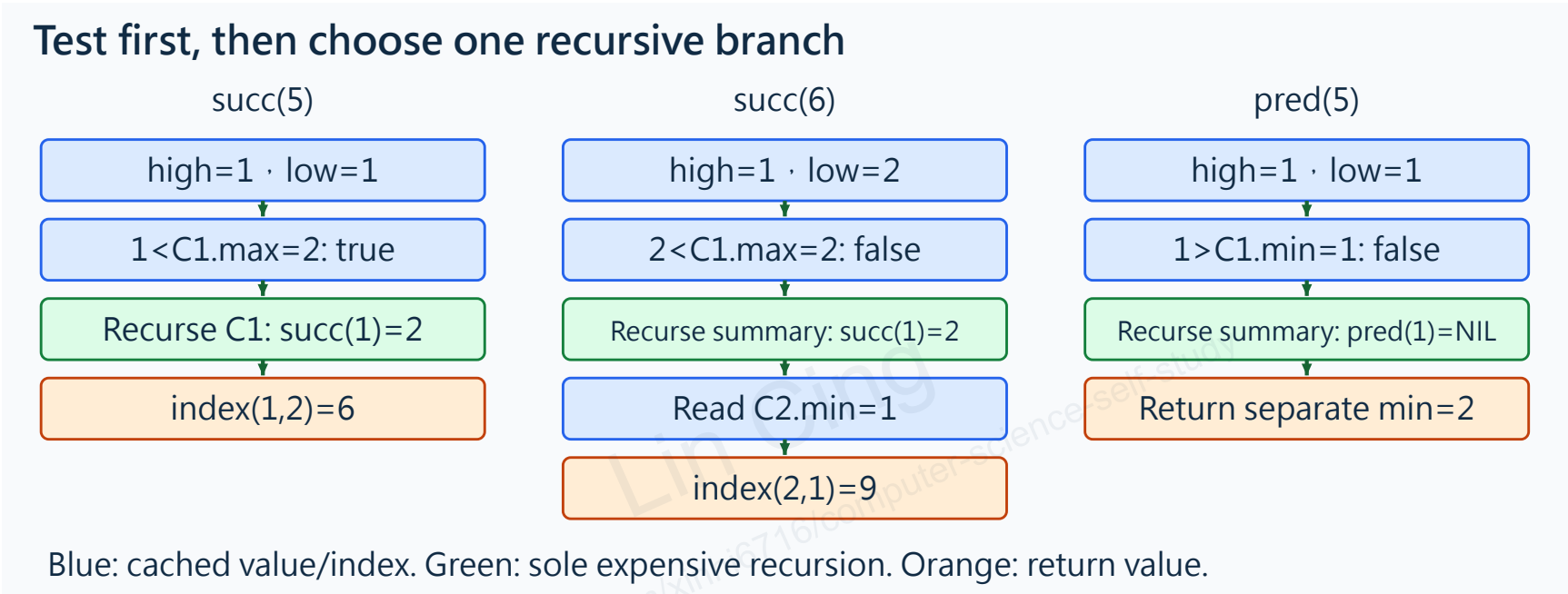
```
predecessor(V,x):
    if V.min==NIL or x<=V.min: return NIL
    if x>V.max: return V.max
    if V.u==2: return 0
    h=high(x); l=low(x); C=V.cluster[h]
    if C.min!=NIL and l>C.min:
        return index(h,predecessor(C,l))
    j=predecessor(V.summary,h)
    if j==NIL: return V.min
    return index(j,V.cluster[j].max)
```

At j=NIL, earlier tests established x>V.min, so global min is a valid answer. Do not mirror successor's NIL branch mechanically: min was deliberately excluded from clusters.

For S={2,5,6,9,12,15}, pred5 has no smaller local key in cluster1 and no preceding summary group, but must still return2.

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Actual same-group and cross-group queries



succ5: high1,low1,cluster1.max2. Since1<2, recurse only into cluster1, get local2, and return1 • 4+2=6.

succ6: high1,low2 is not below cluster1.max2, so skip that cluster. summary.succ1=2; read cluster2.min1 and return2 • 4+1=9.

Actual same-group and cross-group queries (continued)

Queries need not belong to S. succ3 crosses from empty cluster0 to cluster1.min1 and returns5. succ15=NIL and pred2=NIL.

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Insertion swaps min and treats empty groups specially

```
insert-new(V,x):
    if V.min==NIL: empty-insert(V,x); return
    if x<V.min: swap(x,V.min)
    if V.u>2:
        h=high(x); l=low(x); C=V.cluster[h]
        if C.min==NIL:
            insert-new(V.summary,h)
            empty-insert(C,l)
        else: insert-new(C,l)
    if x>V.max: V.max=x
```

A smaller new key stays in min; the old minimum becomes the value inserted into a cluster. Empty-group insertion updates two places, but cluster empty-insert is O(1), leaving only summary expensive. A nonempty group recurses only into its cluster without summary changes.

Insert1 into the example: min becomes1, old min2 enters cluster0 as local2, and summary gains0. Final max remains15. Do not also put new min1 into cluster0.

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Deletion bases and promotion of a new minimum

```
delete-present(V,x):
    if V.min==V.max: V.min=V.max=NIL; return
    if V.u==2:
        V.min=V.max=(1 if x==0 else 0); return
    if x==V.min:
        h=V.summary.min
        x=index(h,V.cluster[h].min)
        V.min=x
    h=high(x); l=low(x); C=V.cluster[h]
    delete-present(C,l)
```

Continue with the next slide's summary/max cleanup. In a multi-element set, deleting min promotes the smallest clustered key, then removes that promoted value from its old cluster to preserve exclusion. Variable x now denotes the promoted value, not the caller's original deleted minimum.

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Complete summary and maximum cleanup

After the preceding deletion in C:

```
if C.min==NIL:
    delete-present(V.summary,h)
    if x==V.max:
        j=V.summary.max
        if j==NIL: V.max=V.min
        else: V.max=index(j,V.cluster[j].max)
elif x==V.max:
    V.max=index(h,C.max)
```

An emptied C had only one element before deletion, so its deletion used O(1) base work before summary recursion. If cluster deletion really recursed through several levels, C cannot become empty and no summary recursion follows. The two expensive branches remain exclusive.

A stale summary can direct later queries into empty groups. A stale max can make successor choose an incorrect branch. These fields are required invariants, not optional hints.

Recompute updates from the same data

Starting at S={2,5,6,9,12,15}:

After	min	max	cluster0	cluster1	cluster2	cluster3	summary
Initial	2	15	Empty	{1,2}	{1}	{0,3}	{1,2,3}
insert1	1	15	{2}	{1,2}	{1}	{0,3}	{0,1,2,3}
delete1	2	15	Empty	{1,2}	{1}	{0,3}	{1,2,3}
delete15	2	12	Empty	{1,2}	{1}	{0}	{1,2,3}

Deleting1 promotes2 from cluster0, then deletes local2 and removes newly empty group0 from summary. Deleting15 leaves local0 in cluster3, so summary retains3 and new max=3・4+0=12. Combining min with all clusters reconstructs every row's complete set without lost or duplicate promoted values.

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Correctness and doubly logarithmic time

Induct on b. At u=2, min/max represent empty, singleton, or{0,1} exactly. Assume operations correct on smaller universes. Distinct high ranges partition values, index inverts high/low, and min is stored separately.

Membership enters its only possible group. Successor/predecessor use local extrema to test same-group existence, otherwise summary finds the nearest group, with separate min handling. Insertion swaps and deletion promotions transfer exactly one representative; summary stays equivalent to nonempty clusters. Thus representation and set semantics are preserved.

Each operation has at most one nonconstant subcall on at most ceil(b/2) bits:

$T(2^b) \le T(2^{\lceil b/2 \rceil}) + O(1)$.

For $b \ge 2$, $\lceil b/2 \rceil \le 2b/3$, so reaching b=1 needs O(log b) levels, giving O(1+log log u). This depends on the proved exclusive branches, not only structural depth.

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Preallocated space recurrence

The original version builds every summary/cluster in advance. Even an empty set uses Θ(u) words and Θ(u) initialization time. Including node fields and cluster-pointer arrays, for u>2,

$$P(2^b)=2^{\lceil b/2\rceil}P(2^{\lfloor b/2\rfloor})+P(2^{\lceil b/2\rceil})+\Theta(2^{\lceil b/2\rceil}).$$

Let F(b)=P(2^b)/2^b and ε_b=2^(−floor(b/2)). For a fixed c,

$$F(b)\le F(\lfloor b/2\rfloor)+\varepsilon_b\ F(\lceil b/2\rceil)+c\ \varepsilon_b.$$

With G=F+c,

$$G(b)\le(1+\varepsilon_b)\max(G(\lfloor b/2\rfloor),G(\lceil b/2\rceil)).$$

Follow the larger term down to b=1 and bound the accumulated finite product, retaining summary cost rather than ignoring it.

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Linear space and allocation limitations

Counting r levels backward along that halving path, bit sizes are at least2^(r−1)+1. Corresponding ε values are bounded by1/2,1/2,1/4,1/16,…,whose sum is bounded independently of b. Since1+x≤e^x,

$$\prod(1+\varepsilon)\le \exp(\sum\varepsilon)=O(1).$$

Thus F(b)=O(1),P(u)=O(u). Cluster branches alone partition the original universe into u/2 base nodes at u=2, giving Ω(u), hence Θ(u). Initializing these fields also costs Θ(u).

For huge u and small n, preallocation can be unattractive. On-demand clusters and hash dictionaries require separate proofs of construction,dictionary time,and space. Calling allocation lazy does not establish O(n) space and worst-case O(log log u) operations simultaneously. The proved version here uses preallocated arrays and direct indexing on word-RAM.

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Sources

Bibliography

Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*, 3rd ed., MIT Press, 2009, Chapter20.

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References and attribution

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